

8 Mutations can give rise to cancer or genetic disorders.

- (a) Cancer is one of the main causes of death in the world.

In 2018, in the UK, there were 541 589 deaths in total and 166 800 of these were due to cancer.

Calculate the ratio of deaths caused by cancer to deaths not caused by cancer.

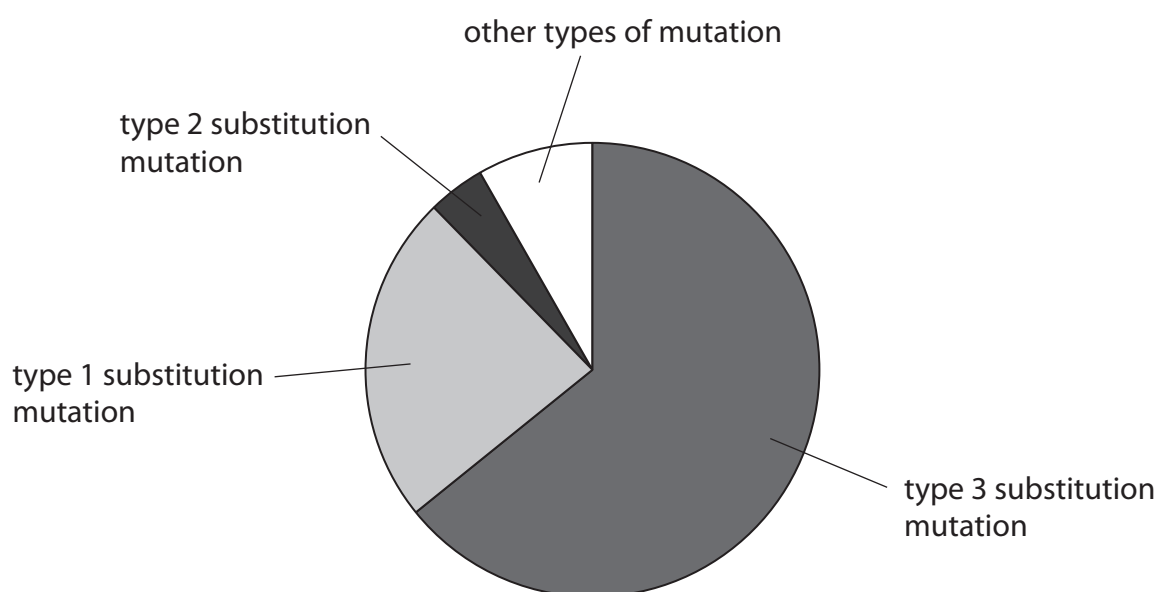
Give your answer to two decimal places.

(2)

Answer

- (b) Scientists have identified many mutations in cancer cells and are trying to identify the mutations that are significant.

The chart shows the proportion of cancers caused by some types of mutation.



- (i) Name **two** other types of mutation.

(1)

1

2

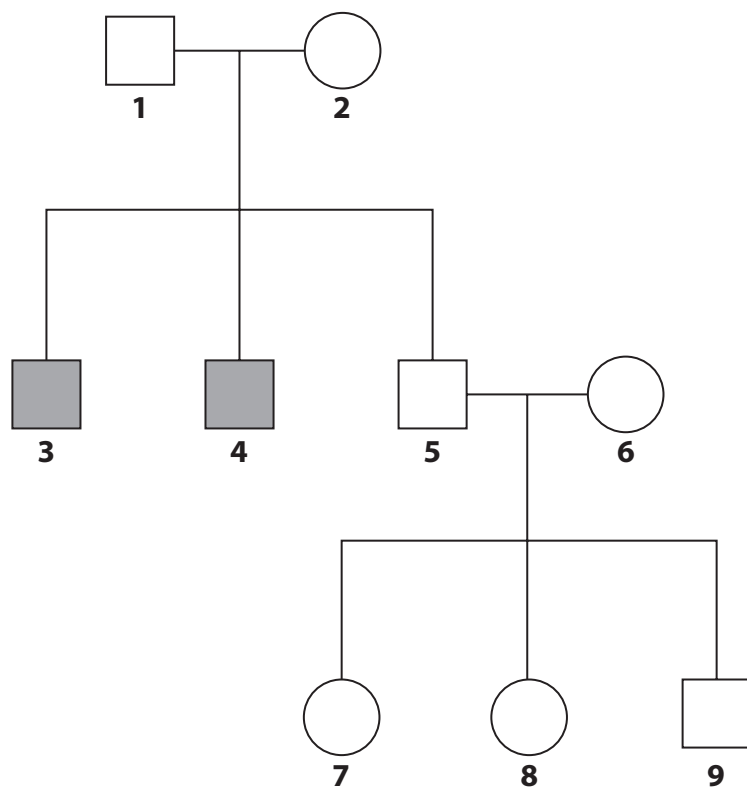


[illegible]

(c) Phenylketonuria (PKU) is a genetic disorder.

This disorder is inherited in a similar way to cystic fibrosis.

The pedigree diagram shows the phenotypes of individuals in one family affected by PKU.



Key:

-  female unaffected by PKU
-  male unaffected by PKU
-  female with PKU
-  male with PKU

(i) What is the probability of couple **1** and **2** having a fourth child that is a male affected by PKU?

(1)

- ☐ **A** 0.000
- ☐ **B** 0.125
- ☐ **C** 0.250
- ☐ **D** 0.500



- *(ii) Discuss the extent to which this pedigree diagram can be used to identify the genotypes of all the members of this family.

(6)

(Total for Question 8 = 15 marks)

TOTAL FOR PAPER = 80 MARKS



P 7 0 9 6 0 A 0 2 7 2 8